Question 1

Planning (15 marks)

Defining the problem (3 marks)

P1 Vary *d* and measure *y* or *d* is the independent variable and *y* is the dependent variable [1]

P2 Keep current constant [1]

P3 Keep length of wire constant [1]

Methods of data collection (5 marks)

M1 Diagram showing ruler positioned and power supply connected to wire or diagram showing initial and final marks on screen and power supply connected to wire [1]

M2 Use of ammeter to check current – penalise incorrect circuit diagrams [1]

M3 Measurement of *d* using micrometer [1]

M4 Allow time for displacement of wire to stabilise [1]

M5 Detail on measuring *y*; final reading - initial reading [1]

Method of analysis (2 marks)

A1 Plot a graph of log *y* against log *d* [1]

A2 *q* = gradient [1]

Safety considerations (1 mark)

S Safety related to hot wire – use of gloves, wait to cool down/switch off before changing wire, do not touch hot wire [1]

Additional detail (4 marks)

D Relevant points might include [4]

1. Use of vernier scale to measure *y* /well described optical method/use of set square
2. Method for keeping current constant e.g. use of rheostat
3. Check starting position for *y* for same wire
4. lg *y* = *q* lg *d* + lg *p*
5. Repeat measurements of *d* at different points along the wire and determine average
6. Control of additional variables e.g. separation between supports, room temperature
7. Use of protective resistor (either labelled or explained).

15 marks can be scored in total.

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